

Finding the Missing Factor

Answer Key

Grade 3 — Multiplication and Division Fact Fluency

Quick Answers

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|------|-----------------------------------|
| 1. 6 | 6. 7 boxes |
| 2. 7 | 7. 9 |
| 3. 6 | 8. 9 trays |
| 4. 7 | 9. 9 |
| 5. 6 | 10. (a) chairs in each row = 8, — |
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3 — Multiplication and Division Fact Fluency

3.OA.A.3 – *problems 6, 8, 10*

3.OA.A.4 – *problems 1, 2, 3*

3.OA.C.7 – *problems 4, 5, 7, 9*

Difficulty 1–5 of 5 across 11 tagged checks.

Common wrong answers

1. *If they got 96: multiplied the total by the number of rows instead of dividing*
 3. *If they got 15: subtracted the row size from the total instead of dividing*
 6. *If they got 448: multiplied the two given numbers instead of dividing*
 8. *If they got 40: subtracted instead of dividing*
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1. A sticker sheet has 4 equal rows and 24 stickers in all. How many stickers are in each row? $4 \times \underline{\quad} = 24$

Solution: The 24 stickers are shared equally among 4 rows, so the missing factor is the size of one row: $24 \div 4$. Skip-count by 4 until you reach 24: 4, 8, 12, 16, 20, 24 — that is six jumps, so $4 \times 6 = 24$. 6

2. A garden has 5 equal rows of seedlings and 35 seedlings in all. How many seedlings are in each row? $5 \times \underline{\quad} = 35$

Solution: One row is $35 \div 5$. Skip-count by 5: 5, 10, 15, 20, 25, 30, 35 — seven jumps. Check with the multiplication fact: $5 \times 7 = 35$. 7

3. Tiles are laid in equal rows of 3 tiles, 18 tiles in all. How many rows are there? _____ $\times$ 3 = 18

Solution: Here the missing factor is the NUMBER OF ROWS, not the row size. Each row uses 3 tiles, so ask how many 3s are in 18: $18 \div 3$. Count by threes: 3, 6, 9, 12, 15, 18 — six threes, so $6 \times 3 = 18$. 6

4. Find the missing factor. $6 \times$ _____ $= 42$

Solution: Ask “six times what gives 42?” The related division fact is $42 \div 6$. From the sixes: $6 \times 6 = 36$ and $6 \times 7 = 42$. 7

5. Find the missing factor. _____ $\times 9 = 54$

Solution: The missing factor is the first one this time, but multiplication can be read in either order: what times 9 makes 54? The related division fact is $54 \div 9$. From the nines: $9 \times 6 = 54$. 6

6. A craft box holds exactly 8 crayons. A classroom has 56 crayons in full boxes. How many boxes? $8 \times$ _____ $= 56$

Solution: Each box is a group of 8, so the missing factor counts the groups: $56 \div 8$. From the eights: $8 \times 7 = 56$. The unit is boxes, because the missing factor counts boxes, not crayons. 7 boxes

7. Find the missing factor. $7 \times$ _____ $= 63$

Solution: Seven times what makes 63? The related division fact is $63 \div 7$. From the sevens: $7 \times 9 = 63$. 9

8. A baking tray holds exactly 5 muffins. A bakery has 45 muffins on full trays. How many trays? $5 \times$ _____ $= 45$

Solution: Each tray is a group of 5, so the missing factor counts the trays: $45 \div 5$. Count by fives to 45 — nine fives — or use $5 \times 9 = 45$. 9 trays

9. Find the missing factor. _____ $\times 8 = 72$

Solution: What times 8 makes 72? The related division fact is $72 \div 8$. The eights run 8, 16, 24, 32, 40, 48, 56, 64, 72 — nine eights, and $9 \times 8 = 72$. 9

10. A class sets up 6 equal rows of chairs, 48 chairs in all. (a) How many chairs are in each row? (b) Explain how a multiplication fact you already know helps you find that missing number.

Solution (a): The 48 chairs are shared equally among 6 rows, so one row is $48 \div 6$. From the sixes: $6 \times 8 = 48$. 8

Solution (b) — instructor-judged. A full-credit answer names a fact the student already knows and says how it produces the missing number. Model answer: “I know $6 \times 8 = 48$, and the equation is $6 \times \square = 48$, so the missing number must be 8.” Skip-counting is equally acceptable: “I counted by sixes — 6, 12, 18, 24, 30, 36, 42, 48 — and that took 8 sixes.” Answers that give only the number 8 with no fact and no count do not earn full credit; answers that say “I divided 48 by 6” earn full credit if the student also states the fact or the count that gave the quotient.